

National Exam of Higher National Diploma 2026 Session

Specialty/option: SOFTWARE ENGINEERING (SWE)

Paper : Case Study

Duration : 6 hours

Credits : 14

GENERAL INSTRUCTIONS.

- **ANSWER ALL QUESTIONS**
- *You are reminded of the necessity of good English and orderly presentation of your material.*

SECTION A: ALGORITHM AND PROGRAMMING. (50 MARKS)

Question 1 (10 Marks)

Reading the fuel consumption in litre according to the table below:

display of the fuel consumption

Reading[litre]	Litre less than or equal to	100 less than litre and less than	Litre greater than
	100	500	500
price	7000 F CFA	50000 F CFA	85000 F CFA

Work Required:

- Draw the flowchart to calculate and print the bill price given the table above. (5 marks)
- Implement the flowchart in the programming language of your choice (5 marks)

Question 2 (20 Marks)

A labor contractor in Cameroon pays the workers at the end of each week according to the rules given below:

For the first 35 hours of work, the rate of pay is CFA7500 per hour; for the next 25 hours, the rate of pay is CFA 9000 per hour; for the rest, the rate of pay is CFA 15000 per hour. No worker is allowed to work for more than 80 hours in a week.

Work Required:

- Develop a stepwise form technique to show how the wages of the workers can be calculated on the basis of valid inputs. (10 marks)
- Implement the stepwise form technique in a programming language of your choice. (10 marks)

Question3 (20 Marks)

A cube is a box-shaped solid in which all sides are of equal length. The volume of a cube is obtained by taking the third power of the length of a side, $\text{Volume} = (\text{Side})^3$

To define the class Cube, we use the following pseudocode:

```
1 Class Cube
2   //attributes of type Cube
3   Declare Side As Float
4   Declare Volume As Float
5   //methods of type Cube
6   Subprogram SetSide(Float NewSide)
7     Set Side = NewSide
8   End Subprogram
9   Subprogram ComputeVolume()
10    Set Volume = Side^3
11  End Subprogram
12  Function GetVolume() As Float
13    Set GetVolume = Volume
14  End Function
15  Function GetSide() As Float
16    Set GetSide = Side
17  End Function
18 End Class
```

Work Required:

- a) Identify the member data of the class Cube (4 marks)
- b) Identify the member functions of the class Cube (4 marks)
- c) Implement two Constructors: constructor without arguments and constructor with arguments (4 marks)
- d) Provide the class definition of the class Cube (2 marks)
- e) Provide the class implementation of the class Cube (2 marks)
- f) Write a test class called CubeTest to test the functionality of the class Cube in C++ (4 marks)

SECTION B: DATABASE DEVELOPMENT AND ADMINISTRATION (20 MARKS)

Question 4 (20 Marks)

- a) Think of an organizational database in which some of the fields in the CUSTOMER table must have the given data types. Explain what they mean and how they are used:
 - i) Customer ID (auto numeric field)
 - ii) Customer Name (text field)
 - iii) Fee Paid (decimal field)

iv) Pay Date (date field).

(2 x 4 = 8 marks)

- b) The table below contains sample data for vehicles and for operators who ply these vehicles. In discussing these data with users, we find that vehicle ID (but not descriptions) uniquely identify vehicles and that operator names uniquely identify operators.

VehicleID	Description	Operator	Route	Tariff Per Mile
V1	Luxury	Polax	Grand Trail	100
		Ubet	East Route	150
V2	Comfort	Polax	Grand Trail	45
		Ubet	East Route	60
		Minim	South Trunk	35

- Convert this table to a relation (named VEHICLE OPERATOR) in first normal form. Illustrate the relation with the sample data in the table. (2 marks)
- List the functional dependencies in VEHICLE OPERATOR and identify a candidate key. (3 marks)
- For the relation VEHICLE OPERATOR, identify each of the following: an insert anomaly, a delete anomaly, and a modification anomaly. (3 marks)
- Draw a relational schema for VEHICLE OPERATOR and show the functional dependencies. (3 marks)
- In what normal form is this relation? (2 mark)

SECTION C: WEB DESIGN (15 MARKS)

Question 1 (10 Marks)

You are to design a menu for the "FEED WELL RESTAURANT". The menu should be presented exactly as in the Table 2 below.

Table 2: Menu for Feed Well Restaurant:

Days	MENU OF THE DAY		Cost/FCFA
	Breakfast	Luanch	
MONDAY	Achombo	Khati-khati	2,750
TUESDAY	Pizza pie	Esuba	3,500
WEDNESDAY	Sandwich	Orebetem	5,700

Work Required:

- a) Use the following code fragment inline:

```
<style type="text/css">
  td, th{border:1px solid #ccc;}
  table{border: 1px solid black}
</style>
```

OR the following values: table width = 450, borders = 1

- b) Use #F7F7F7 for the table background colour
 - c) Use the *Caption element* to add the title over the table.
 - d) Align days to the left, and Cost/FCFA values to the right.
- By using a suitable web technology, design the above table.**

(15 marks)

SECTION D NETWORKING (15 MARKS)

Question 1 (15 Marks)

- a) The process in which two or more computers or devices transfer, data, instructions and information is known as digital communication. Select from the following terms the ones you need to have successful communication: tablets, sending device, receiving device, computer, a communication device, transmission media. (3 marks)
- b) Name at least four advantages of a network. (3 marks)
- c) In the following terms are standards and protocols used in network communication. Sort them into name, type and sample usage in a table that you will draw with three columns; Ethernet, LTE, WiFi, Bluetooth, Token ring, TCP/IP. (3 marks)
- d) You are attempting to use Bluetooth to send files from your phone to your computer. When you try sending the files from your phone, it does not display your computer as a device to which it can transfer the file. What might be the problem? (3 marks)
- e) What steps are involved in maintaining a network? (3 marks)